

BRAC University
Department of Computer Science and Engineering
CSE 460, Section 04, Fall 2025

Name:

Student ID:

Q1. Lee's Maze algorithm can be used to find the shortest path between two nodes avoiding obstacles.

Dark regions are metals, **X** are obstacles or components.

T2	X					
				X		
X						T1
					X	
	X					
		S	X			X
					X	

Find the shortest path from source, S to both the targets, T1 and T2. Rule of thumb should be taken into consideration.

Q2. Suppose you have a global bounding box of size 20*20. The size of each grid is 0.5*0.5. You have 3 blocks with the following areas-

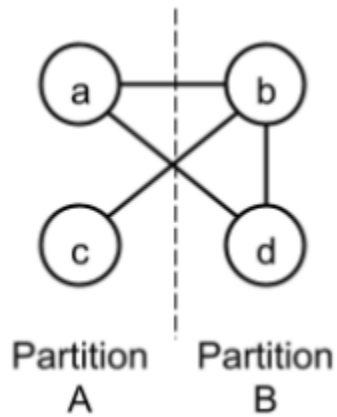
Block A = 5

Block B = 1

Block C = 3

Draw the optimum floor plan.

Q3. The graph below (nodes a-d) can be optimally partitioned using the Kernighan-Lin algorithm. The dotted line represents the initial partitioning. Assume all the edges have the same weight.



- i. Calculate the initial cut cost.
- ii. Identify how many iterations are needed in a single pass?
- iii. Perform the first pass of the algorithm and determine the new cut cost of the optimum output of the first pass.